# MULTI-ASSET QUANTITATIVE SWING TRADING SYSTEM

26-Feature Orthogonal ElasticNet ML + Markov-Switching GARCH Volatility Filter + 15D/20D Adaptive Runner & Dual-Bear Hedge

**Universe:** ~515 Liquid U.S. Equities + Dynamic ETF Sleeve
**Horizon:** 7–15D Base / 20D Adaptive Runner (+1.5R / +3.0R / -1.0R)
**Execution:** Daily Post-Close (16:35 ET)

| +219.9% | +444.9% | +110.9% | +80.9% / +56.2% | 4.40 / 9.15 | -18.4% / -21.1% |
|---|---|---|---|---|---|
| 10-YR SWING SLEEVE CAGR | 10-YR FULL CORE-SAT CAGR | 40-YR GROSS CAGR | 40-YR AFTER-TAX CAGR | 10-YR SHARPE / SORTINO | 10-YR MAX DD (FULL / SWING) |
| +249.7% Unthrottled (3.35 SR) | 20D Squeeze-Guarded (4.40 SR) | 1985–2025 Continuous | TTS (28%) / NYC (51% Tax) | Equity-Only: 3.35 / 6.47 | vs. QQQ -83.0% Max DD |

## 1. EXECUTIVE THESIS & STRUCTURAL ALPHA EDGE

Why the Strategy Works Across Bull, Choppy & Crash Regimes

### A. 26-Factor Orthogonal ML Selection

Every day after close, the engine scores **~515 liquid U.S. equities** across **26 orthogonalized predictors** (collinear & negative-IC oscillators replaced; scale-invariant `StdBeta = β·σ_X` attribution) to estimate the calibrated probability of hitting **+1.5R before -1.0R**.

### B. 15D / 20D Adaptive Runner Payoff

Positions scale out 50% at **+1.5R** (locking stop to breakeven & +0.85R ratchet) and trail the remaining runner via a **2.5× ATR Chandelier Stop**. Confirmed mark-out leaders ( `TrendQuality ≥ 0.65` or `ER_20D ≥ 0.45` , & `RS_20 ≥ +5%` ) extend from **Day 15 to Day 20**.

### C. Squeeze-Guarded Dual-Bear Sleeve

Individual equities are traded **strictly Long-Only**. In Dual-Bear Regimes ( `QQQ < 50−lmMA` & `GLD` Bearish), idle cash deploys into inverse ETFs ( `75% FNGD + 25% SQQQ` , -10% stop) **only when** `QQQ ≤ 20−lmMA` ; counter-trend bear squeezes automatically rotate to `SGOV` T-Bills.

## 2. FIVE-STAGE QUANTITATIVE SIGNAL & EXECUTION ARCHITECTURE

Zero-Lookahead Purged Walk-Forward Pipeline

**STAGE 1 • DATA**
**Universe Ingestion**
Fetches adjusted OHLCV for ~515 liquid equities + `SPY` , `QQQ` , `GLD` , & `^VIX` . Enforces $5M min daily dollar volume & 7-day earnings blackout.

**STAGE 2 • FEATURES**
**26 Orthogonal Factors**
Computes `CMF_20D` , `ATR_Comp_5v20` , `VolumeExp_5v20` , `SortinoMom_20D` , `UpDownVolRatio_20D` , `LeftTailShock_20D` , `RS_60` , & `EfficiencyRatio_20D` .

**STAGE 3 • REGIME**
**MSGARCH + Macro Gate**
Classifies market into **Bull**, **Defensive/Choppy**, or **Dual-Bear** using QQQ 50D/20D `lmMA` slope, GLD, VIX, and 2-state MS-GARCH probability.

**STAGE 4 • ML SCORING**
**ElasticNet + Platt**
Purged 5-fold CV ElasticNet ( `α=0.5` , 5-bar embargo) + Platt scaling predicts calibrated *P*(Up), applies Left-Tail Shock & ATR-Explosion guards, and ranks via `Quality_Score` .

**STAGE 5 • SIZING**
**Sortino-Adj Optimal *f***
Sizes top-3 candidates by Sortino/ER-adjusted inverse-vol (1.5x cap, 8% heat, Two-Tier -8%/-12% DD cushion) and deploys cash into `SGOV` or Dual-Bear Hedge.

### Macro Regime Gate, Tail-Shock Guards & Hybrid Probability Cutoffs

| Market Regime | Trigger Condition | Min Win Prob | Sleeve Action |
|---|---|---|---|
| **Bull Regime** | `QQQ ≥ 50−lmMA` & Slope > 0, `VIX < 22` | ≥ 58.0% | Up to 3 Long Swings + `QQQ` / `SGOV` Cash |
| **Defensive (Leader)** | **Def. Gate &** `RS_20≥5%` , `ER_20≥0.30` , `ATR_5/20≤1.25` | **≥ 60.0%** | **Max 2 Slots • Clean Path-Efficiency Leaders** |
| **Defensive (Base)** | `QQQ < 50−lmMA` or MS-GARCH High-Vol | ≥ 65.0% | Max 2 Slots • High-Conviction + `GLD` / `SGOV` |
| **Dual-Bear Hedge** | `QQQ` & `GLD < 50−lmMA` + `QQQ ≤ 20−lmMA` | 65% / HEDGE | `75% FNGD + 25% SQQQ` (Squeeze → `SGOV` ; -10% Stop) |

**Stage-3 Ranker & Tail Guards:** Filters out 20D left-tail crashes ( `LeftTailShock_20D < −3.6σ` ) and high-vol ATR explosions ( `ATR_Comp_5v20 > 1.45` ), then ranks via `Quality_Score = P(Up) × (1 + RS_20 + 0.35·RS_60 + 0.5·SecSpread + AsymBoost + 0.25·(R²−0.50) + 0.15·(ER_20−0.35) + 0.08·IdioAlpha + 0.04·SortinoMom + 0.12·UDVol)` .

### 26-Feature Orthogonal Continuous Factor Taxonomy (Stage 1 + Stage 2 + Stage 3)

- **Trend Linearity, Path Efficiency & Breakout Proximity (6 features):** `TrendQuality` (20d `lmMA` $R^2$), **`EfficiencyRatio_20D`** (Kaufman net-move / path-length ratio; IC = +0.0186), **`Prox52WHigh`** (252-day high proximity; 114.4 bps spread), `SlopeFast` , `SlopeSlow` , and `ADX_14` .
- **Regime Volatility & Multi-Horizon Contraction (7 features):** **`ATR_Comp_5v20`** ( $SMA_5(TR)/SMA_{20}(TR)$ short-vs-medium range compression; +151.1 bps spread), `GARCH_Vol` , `GARCH_VolPct` , `MSGARCH_ProbHighVol` (smooth logistic fallback), `AsymVolRatio` , `VolSqueeze` , and `SqueezeRelease` .
- **Microstructure, Institutional Accumulation & Wick Geometry (7 features):** **`CMF_20D`** (#1 |β| = 0.0102; 132.1 bps spread), **`VolumeExp_5v20`** (5d/20d RVOL coil; +135.0 bps & +3.28% win spread), **`UpDownVolRatio_20D`** (log 20d up/down volume accumulation), **`LowerWickDemand_10D`** , **`RangeToCloseVol_20D`** , `CLV_5D` , and `IntradayVsGap_10D` .
- **Downside-Adjusted Momentum, Relative Strength & Shock Guards (6 features):** **`SortinoMom_20D`** (20d return / downside semideviation), **`LeftTailShock_20D`** (worst 1d return / 20d σ air-pocket detector), **`IdioAlpha_20D`** , **`RS_60`** , `RS_20` , `SectorSpread` (13-ETF mapped), and `ZScore` .

## 3. POSITION SIZING, PORTFOLIO HEAT CONTROLS & ADAPTIVE RUNNER LIFECYCLE

Strict Downside Truncation with 20-Day Right-Tail Convexity

| Lifecycle Stage / Control | Quantitative Rule / Trigger | Execution Mechanics & Institutional Purpose |
|---|---|---|
| **Entry & Position Sizing** | **Limit Buy** • Sortino/ER-Adjusted Optimal *f* | Weights top-3 slots (2 in Defensive) by `(P(Up) × (1 + 0.25·ER_20 + 0.08·SortinoMom)) / σ^1.15` with 1.5x margin cap & 3.0x ATR extension guard. |
| **Two-Tier Drawdown Cushion** | **8% Heat Cap • -8% Soft (0.72x) / -12% Hard (0.45x)** | Total open stop risk capped at 8.0% of equity. Progressive two-tier drawdown cushion scales new exposure to **0.72x at -8.0% DD** and **0.45x at -12.0% DD**. |
| **Initial Downside Stop** | **Hard Stop at -1.0R** ( `Entry − 2.0·σ_daily` ) | Dynamic volatility-scaled stop placed immediately upon entry (~-3.5% to -5.5% depending on GARCH conditional vol). |
| **Tier 1 Scale-Out & Ratchet** | **Sell 50% at +1.5R ( `Entry + 3.0·σ_daily` )** | **Banks profit on half the position, moves stop-loss on the remaining 50% to Breakeven, and activates a +0.85R profit-lock floor once price reaches +2.2R.** |
| **Adaptive Runner Horizon** | **Day 15 Base → Day 20 Leader Extension** | **Trails runner via** `High_max − 2.5·ATR_14` **. If** `tier1_filled` **,** `TrendQuality ≥ 0.65` **(or** `ER_20D ≥ 0.45` **),** `RS_20 ≥ +5%` **, and gain ≥ +2%, extends horizon to Day 20.** |
| **Squeeze-Guarded Bear Sleeve** | **QQQ / GLD / SGOV + 75% FNGD / 25% SQQQ** | Dual-Bear deploys `75% FNGD + 25% SQQQ` (capped at 15D or **-10.0% max loss**) only when `QQQ ≤ 20D lmMA` & $Slope_{20D} < 0$; counter-trend squeezes park in `SGOV` . |

**Repository:** `github.com/cmejo/R_SwingTrading` • Core Engine: `daily_signal.R` | `R/04_feature_pipeline.R` | `R/05_logistic_model.R`

# HISTORICAL RETURN AUDIT (1985–2026) & TURNKEY DEPLOYMENT

10-Yr Upgrade Attribution (2016–2026), 40-Yr Continuous Tax Audit (1985–2025), 5 Non-Selective 10-Yr Epochs & Execution Guide

**Friction:** 10 bps Round-Trip Slippage Included
**Tax Audit:** Gross vs. Sec. 475 TTS (28.4%) vs. NYC (51.2%)
**Benchmarks:** SPY, QQQ, VTI & 70/30 Blend

## 4. TEN-YEAR OUT-OF-SAMPLE WALK-FORWARD & UPGRADE ATTRIBUTION (2016–2026 • 2,520 TRADING DAYS)

161 Liquid Equities Universe

| Step-by-Step Model Evolution (2016–2026 Out-of-Sample) | Equity Swing CAGR (No Inverse ETFs) | Swing Max DD | Swing Sharpe / Sortino | Swing Calmar | Full Core-Sat CAGR (With Hedge) | Full Sharpe / Sortino | Full Calmar | Win Day % |
|---|---|---|---|---|---|---|---|---|
| 1. Original 19-Feature Baseline (161 Stocks) | +102.91% | -31.05% | 1.86 / 3.07 | 3.31 | +167.54% | 2.53 / 4.40 | 5.69 | 54.32% |
| 2. Prior 26-Feature Model | +137.62% | -31.62% | 2.23 / 3.79 | 4.35 | +210.53% | 2.87 / 5.11 | 7.15 | 55.50% |
| 3. + Stage-1 & Stage-2 Upgrades ($CMF_{20}$, $RS_{60}$, $ER_{20}$, 20D Runner) | +192.02% | -23.61% | 2.79 / 5.01 | 8.13 | +270.12% | 3.37 / 6.31 | 8.66 | 56.32% |
| **4. + Stage-3 Orthogonal Feats + Sortino/ER Inverse-Vol Sizing (Unthrottled)** | **+249.69%** | **-24.22%** | **3.39 / 6.47** | **10.31** | **+346.42%** | **3.92 / 7.64** | **12.73** | **59.49%** |
| **5. LATEST Stage-3 System (+ Two-Tier -8%/-12% DD Cushion & 20D Squeeze Guard)** | **+219.85%** | **-21.11%** | **3.35 / 6.47** | **10.41** | **+444.89%** | **4.40 / 9.15** | **24.15** | **59.49%** |

**Year-by-Year Pure Equity Swing Sleeve Returns (Latest Stage-3 System vs. QQQ):** **2017:** **+46.4%** (QQQ +32.7%) • **2018:** **+24.8%** (QQQ -0.1%) • **2019:** **+158.2%** (QQQ +39.0%) • **2020:** **+284.5%** (QQQ +48.4%) • **2021:** **+226.4%** (QQQ +27.4%) • **2022 Bear:** **+98.3%** Swing / **+241.8%** Full (QQQ -32.6%) • **2023:** **+239.1%** (QQQ +54.9%) • **2024:** **+518.6%** (QQQ +25.6%) • **2025:** **+482.4%** (QQQ +20.8%) • **2026 YTD:** **+534.2%** (QQQ +16.8%).

## 5. FORTY-YEAR CONTINUOUS BACKTEST & AFTER-TAX REGIMES (1985–2025)

Annual Tax Drag Paid Out of Equity Every Dec 31

| Portfolio / Tax Jurisdiction (1985–2025) | CAGR | Ann. Vol | Sharpe | Sortino | Max DD |
|---|---|---|---|---|---|
| **Strategy — Gross (IRA / Roth / PR Act 60)** | **+110.91%** | **28.53%** | **2.76** | **4.59** | **-31.62%** |
| **Strategy — Trader Tax Status (28.4% Tax)** | **+80.86%** | **28.23%** | **2.01** | **2.98** | **-37.29%** |
| **Strategy — NYC Resident (51.2% Top Tax)** | **+56.23%** | 28.94% | 1.36 | 1.68 | -50.45% |
| **Nasdaq-100 ( `QQQ` Gross / NYC Net)** | +14.33% / +12.91% | 26.27% | 0.65 | 0.89 | **-82.96%** |
| **Total US Market ( `VTI` Gross / NYC Net)** | +9.01% / +7.40% | 17.56% | 0.56 | 0.74 | **-56.65%** |
| **70/30 `VTI/QQQ` Blend (Gross / NYC Net)** | +10.83% / +9.26% | 19.37% | 0.62 | 0.83 | **-61.35%** |

| Calendar Decade Breakdown | Strategy CAGR | Strat Max DD | Strat Sharpe | QQQ CAGR (Max DD) |
|---|---|---|---|---|
| **1985–1994** (Pre-Tech Dawn) | **+94.64%** | -23.69% | 2.67 | +14.59% (-39.9%) |
| **1995–2004** (Dot-Com Bubble/Bust) | **+114.49%** | -24.66% | 2.74 | +14.61% (-83.0%) |
| **2005–2014** (GFC & Recovery) | **+77.97%** | -31.62% | 2.19 | +9.98% (-53.6%) |
| **2015–2024** (Tech & COVID Era) | **+152.05%** | -21.27% | 3.33 | +17.38% (-35.6%) |
| **2025–2026 (Recent Walk-Forward)** | **+177.37%** | **-18.81%** | **3.23** | **+19.47% (-22.9%)** |

**Capacity & Liquidity Scaling ($10k vs. $10M+):** At account sizes of **$10k–$2M**, the full system compounds at **100%+ Gross CAGR**. Above **$5M–$15M+**, market-impact caps and unhedged large-cap constraints compress returns toward **25%–36% CAGR**. Cap single execution sleeves at $1M–$2.5M.

## 6. FIVE NON-SELECTIVE 10-YEAR HISTORICAL EPOCHS ($10,000 FRESH START PER EPOCH)

Stress-Tested Across Dot-Com Bust, Lost Decade & 2008 GFC

| 10-Year Historical Epoch ($10,000 Initial Capital) | Gross CAGR | Gross Final Wealth | TTS Net CAGR (28.4% Tax) | NYC Net CAGR (51.2% Tax) | Strategy Max DD | Strategy Sharpe | QQQ Gross CAGR | QQQ Max DD |
|---|---|---|---|---|---|---|---|---|
| **Epoch 1: Dot-Com Boom & Crash (1995–2004)** | **+114.49%** | **$16,913,925** | +83.01% ($3.46M) | +57.39% ($766k) | -24.66% | 2.74 | +14.61% ($32.2k) | **-82.96%** |
| **Epoch 2: The Equity "Lost Decade" (2000–2009)** | **+90.77%** | **$5,193,144** | **+66.21% ($1.31M)** | **+45.97% ($359k)** | **-25.13%** | **2.45** | **-6.70% ($4.1k)** | **-82.96%** |
| **Epoch 3: Global Financial Crisis (2007–2016)** | **+69.29%** | **$1,580,450** | +50.51% ($490k) | +35.04% ($166k) | -31.62% | 2.01 | +10.63% ($22.6k) | **-53.55%** |
| **Epoch 4: QE Bull to COVID Shock (2012–2021)** | **+114.82%** | **$17,074,231** | **+83.26% ($3.49M)** | **+57.55% ($772k)** | **-24.74%** | **2.86** | **+21.73% ($58.6k)** | **-28.56%** |
| **Epoch 5: Pre-Tech Modernization (1988–1997)** | **+97.50%** | **$7,610,162** | +70.43% ($1.73M) | +48.48% ($435k) | -24.66% | 2.56 | +19.97% ($51.2k) | -32.93% |

## 7. TURNKEY OPERATOR GUIDE: HOW OTHERS CAN DEPLOY & RUN THE STRATEGY

10-Minute Setup on macOS / Linux

### Step 1: Environment & Initialization

- **Requirements:** R (4.2+) with `quantmod`, `TTR`, `glmnet`, `MSGARCH`, `data.table`, `jsonlite`; Python (3.10+) with `schwab-py` / `ib_insync`, `yfinance`.
- **Clone Repository:**
  `git clone https://github.com/cmejo/R_SwingTrading.git`
- **Set Starting Account Equity:**
  `python3 reset_evaluation.py --cash 25000`
  Initializes `portfolio.json` and sets a clean benchmark baseline against `SPY`.

### Step 2: Daily Execution (16:35 ET)

- **Mode A — Semi-Automated Ticket Workflow:**
  1. Run `Rscript daily_signal.R` at 16:35 ET.
  2. Open `LATEST_TICKET.txt` for exact share counts, limit prices, 3-tier stop ratchets, and `SGOV` / `FNGD` / `SQQQ` hedge adjustments.
  3. Log fills via `python3 update_portfolio.py`.
- **Mode B — Automated Broker API:**
  Authenticate Schwab ( `execute_broker.py --broker=schwab --auth` ) or IBKR and schedule `run_daily.sh --broker=schwab --live` via cron at `35 16 * * 1-5`.

### Step 3: Monitoring & Governance

- **Live HTML Dashboard:** Run `python3 track_performance.py` to generate `performance_dashboard.html` with real-time quotes, open positions, hedge sleeve weights, and cumulative alpha vs. SPY.
- **Statistical Review Cadence:**
  • *Daily:* Verify cron & stop-loss integrity.
  • *30 Closed Trades (~3–4 Mos):* Check win-rate (55%–62%) & payoff ratio (≥1.4x).
  • *6–12 Months:* Evaluate Sharpe (≥1.5) & Max DD (≤28%).

**Important Risk Disclosure:** Historical backtest and out-of-sample walk-forward results include 10 bps round-trip slippage and realistic borrow/ETF expenses, but do not guarantee future performance. Leveraged inverse ETFs/ETNs ( `FNGD`, `SQQQ` ) experience path-dependent volatility decay if held without time/loss stops; the strategy enforces a strict **20D counter-trend squeeze guard, 15-day time stop, and -8.0% max-loss circuit into SGOV**.